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A fractional order numerical study for the influenza disease mathematical model



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Abstract The motive of these investigations is to present the numerical performances of the fractional order mathematical influenza disease model (FO-MIDM) by designing the computational framework based on the stochastic Levenberg-Marquardt backpropagation neural networks (LMBNNs). The fractional order derivatives have been used to get more accurate performances of the MIDM as compared to the integer order. The MIDM is divided into four subcategories, (i) susceptible $S(q)$, (ii) infected $I(q)$, (iii) recovered $R(q)$ and (iv) cross-immune people $C(q)$. Three different cases based FO derivatives have been numerically presented by using the MIDM. The achieved results based on the MIDM have been presented by using the computing stochastic structure LMBNNs through the process of training, confirmation and testing to decrease the mean square error (MSE) values using the reference (data-based) results. To observe the competence, precision, capability and aptitude of the proposed computing structure LMBNNs, a comprehensive investigation is accessible by performing the correlation, MSE, error histograms, information of state transitions and regression analysis. The worth of LMBNNs procedure is validated through the overlapping of the results with good measures up to the accuracy of 5 to 7 decimals for solving the MIDM.

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1. Introduction

Influenza is such a serious disease that is considered dangerous to the older or those individuals, who have any serious disease (lung issues, diabetes, cancer, kidney or heart issues). It mainly affects the upper part of the respiratory system, throat, bronchi and nose [1]. The human lungs have been affected

mostly through the influenza with high or low ratio. The recovery ratio of this disease is very high and most of the infected people get healthier in 10 to 15 days without using any serious precautionary measures or medical treatment. The epidemic ratio of this disease is calculated around at 5 to 15% annually that has been affected through the higher respiratory pollutions. The influenza epidemic ratio is reported in the ranges of 3–5 million, whereas the death ratios have been calculated as 1/4 to 1/2 million [2,3]. There are various epidemiology models signify the autonomous based on the ordinary system of differential equations, which have been used to represent the system parameters (time independent). Based on these sys-

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tems, the variable quantity represents the subpopulations of infected, recovered, susceptible along with the disease spreading vectors (see Tables 1–2).

There are various schemes that have been applied to present the numerical performances of the mathematical form of the influenza disease model (MIDM). Few of them are presented as: Astuti et al [4] provided the theory of differential form based multi-step transformation approach to present the solutions of the MIDM. Alzahrani et al. [5] presented a numerical computational method to solve the fractional kind of MIDM. Erdem et al [6] discussed the procedure of defective quarantine for the MIDM. Sun et al [7] provided a multi-objective optimization scheme based on the patient provision using the parameters of the MIDM. Ghanbari et al [8] studied the coupled system of avian influenza connecting the derivatives of powerful memories along with the Mittag-Leffler fractional-fractal model. González-Parra et al. [9] authenticated a wide spreading fractional kind of model based on the replication of influenza occurrences. Tchuente et al [10] proposed the media coverage effects using the spread of influenza. Schulze-Horsel et al [11] investigated the infection subtleties along with the virus-induced programmed cell death using the vaccine manufacture based culture influenza. Hovav et al [12] studied the net flow distribution system of influenza vaccination using the supply health limitation. Kanyiri et al [13] accessible the submissions of the optimal control to perform

the pneumonia influenza infection using the antiviral treatment conflict. Patel et al [14] observed the vaccination process of the epidemic influenza through the process of genetic algorithm.

The purpose of this work is to present the numerical performances of the fractional order mathematical influenza disease model (FO-MIDM) by designing the computational framework of the stochastic Levenberg-Marquardt backpropagation neural networks (LMBNNs). The artificial intelligence (AI) based stochastic numerical investigations have been applied in various recent applications, like virus spread based plant system [15], mosquito spreading in an assorted atmosphere [16], dynamical investigations of the corneal shape mathematical nonlinear system [17,18], soft materials along with the micro vessels investigations through the responsive transport system [19], spread of HIV contagion model [20–23], coronavirus diagnosis system [24], speech excitement acknowledgment [25], food chain mathematical systems [26,27], functional order models [28], smoking nonlinear models [29] and singular thermal explosion model [30]. Based on these AI based stochastic applications, the authors are inspired to present the numerical simulations of the FO-MIDM. Few novel points of this study are presented as:

- The fractional order study is presented based on the mathematical form of the influenza disease model.
- The Caputo fractional order derivatives have been used to get more realistic solutions to solve the FO-MIDM.
- Three different variants of the FO values (α), which have been taken between 0 and 1 in order to perform the numerical solutions of the FO-MIDM.
- The design of the computing stochastic structure based LMBNNs is presented first time to solve the FO-MIDM numerically.
- The correctness of the computing stochastic LMBNNs structure is observed through the overlapping of the proposed and the reference numerical solutions.
- The relative representations have been presented by using different measures based error histograms (EHs), correlation metrics, regression performances and mean square error (MSE) values, which improve the strength of the computing stochastic structure LMBNNs to solve the FO-MIDM.

The division of the integer order MIDM is presented into four subcategories, i) susceptible $S(q)$, ii) infected $I(q)$, iii) recovered $R(q)$ and iv) cross-immune people $C(q)$ [31]. The graphic illustrations of the MIDM are illustrated in Fig. 1, while the general form of the MIDM is given in Eq. (1).

Table 1 Parameter setting for the LMBNNs implementation.

Index	Settings
Fitness purpose (MSE)	0
Maximum iterations for the process of learning	500
Adaptive parameter (mu)	0.005
Increasing influence for mu	10
Decreasing impacts for mu	0.1
Maximum values of mu	10^{10}
Minimum form of the gradient routine	10^{-07}
Count of authentication fail	6
Hidden neuron	10
validation and testing trials	10%
Testing and training trials	80%
Selection of the samples	Arbitrary
Dataset generation solver	Adam method
Output, hidden and Input layers	Single for all three
Adam solver to stop and implement the parameter	Default

Table 2 MSE measures based stochastic scheme to solve the MIDM.

Variant	MSE			Performance	Gradient	Mu	Iteration	Time
	Testing	Training	Validation					
1	2.7×10^{-09}	9.3×10^{-10}	3.5×10^{-11}	9.36×10^{-10}	9.8×10^{-07}	1×10^{-10}	96	3
2	1.5×10^{-10}	1.5×10^{-09}	1.4×10^{-10}	1.56×10^{-09}	9.4×10^{-08}	1×10^{-10}	66	2
3	4.0×10^{-11}	5.0×10^{-10}	2.4×10^{-10}	5.02×10^{-10}	9.3×10^{-08}	1×10^{-10}	61	2

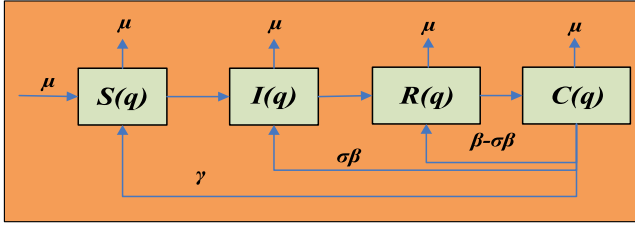


Fig. 1 Graphic illustrations of the MIDM.

$$\begin{cases} \frac{dS(q)}{dq} = \gamma C(q) + \mu - \beta \mu S(q)I(q) - \mu S(q), & S(0) = m_1, \\ \frac{dI(q)}{dq} = \sigma \beta I(q)C(q) - \psi I(q) + \beta I(q)S(q) - \mu I(q), & I(0) = m_2, \\ \frac{dR(q)}{dq} = \beta C(q)I(q) - \beta \sigma C(q)I(q) - (\delta + \mu)R(q) + \psi I(q), & R(0) = m_3, \\ \frac{dC(q)}{dq} = -\gamma C(q) + \delta R(q) - \beta I(q)C(q) - \mu C(q), & C(0) = m_4, \end{cases} \quad (1)$$

where β represents the rate of contact/transmission for the susceptible class to infected category. The exposed form of the cross-immune is indicated by σ based on the time unit into the communicable subpopulation. The initial conditions (ICs) shown in the system (1) are m_1 , m_2 , m_3 and m_4 . The terms based on the transmittable, infected and cross-immune are represented as δ^{-1} , γ^{-1} and α^{-1} , respectively. The essential particulars of the MIDM are signified in the system (1) based on the underlying perceptions along with the essential derivation, theoretical explanation, operating point using the ICs and parameter validation, which are obtainable in the stated investigations for the concerned scholars [32,33].

The rest of the paper is systematized as: Section 2 presents the design of the FO-MIDM. Section 3 shows the stochastic structure based on the LMBNNs. Section 4 provides the details of the results, while the concluding remarks and future research studies are presented in the last Section.

2. Design of the FO-MIDM

The fractional calculus (FC) investigations have been presented centuries ago during the period of Newton, but it has achieved enormous significance in these decades. The idea of the FO derivatives is modernized based on the complexities to the typical inhomogeneity form. The multilayered media structure along with the process of diffusion is presented by using the FC operators. A complex phenomenon examined by many scientists with the broad range of computational approaches by using different software of the integer/FO derivatives [20,23,34-39]. The FO derivatives have also been performed to achieve the numerical or analytical results based on the nonlinear differential models [40-47]. The FO form of the MIDM is presented as:

$$\begin{cases} \frac{d^\alpha S(q)}{dq^\alpha} = \mu - \mu S(q) + \gamma C(q) - \beta \mu S(q)I(q), & S(0) = m_1, \\ \frac{d^\alpha I(q)}{dq^\alpha} = \sigma \beta I(q)C(q) - \psi I(q) + \beta I(q)S(q) - \mu I(q), & I(0) = m_2, \\ \frac{d^\alpha R(q)}{dq^\alpha} = \beta C(q)I(q) - \beta \sigma C(q)I(q) - (\delta + \mu)R(q) + \psi I(q), & R(0) = m_3, \\ \frac{d^\alpha C(q)}{dq^\alpha} = -\gamma C(q) - \beta I(q)C(q) + \delta R(q) - \mu C(q), & C(0) = m_4, \end{cases} \quad (2)$$

where α is the FO derivative using the sense of Caputo derivative to present the FO-MIDM shown in the system (1) and the values of the FO have been selected between [0, 1]. This aim of this work is to solve the FO-MIDM through the procedures of the AI together with the LMBNNs [48,49].

3. Designed stochastic structure: LMBNNs

In this section, the significant performances to establish the computing stochastic LMBNNs structure are presented to solve each variant of the FO-MIDM. Fig. 2 provides the workflow stepwise illustrations based on the computing stochastic scheme to solve the FO-MIDM. The 1st part of Fig. 2 shows the indication of the model, the mathematical depictions of the system are given in the 2nd part, in the 3rd part, the procedural steps of the LMBNNs are given, while the last step of Fig. 2 simulations of the results are described.

The first step shows the dataset based on the different variants of the FO-MIDM generated by the reference (Adam method) solutions to solve the differential model ('NDSolve' command in the Mathematica package). In this command 'Adams' is applied by using the default setting of the parameter for implementation and termination tolerances. The statics is generated for the FO-MIDM by using the comparison of the presentation in the remaining investigations. Additionally, the necessary particulars of the reference solutions can be reported in the references [50-53].

Fig. 3 shows the 2nd part of the stochastic paradigms LMBNNs, which is expressed in the general perception based on the model of single neuron. The 1st part of Fig. 3(a) shows the construction of the multi-layer neural networks, while the 2nd part of Fig. 3(b) presents the dynamics of the FO-MIDM. The designed LMBNNs is implemented in the Matlab command 'nftool' for the appropriate portions of hidden neurons, testing statistics, learning arrangements and verification material. The executing form of the LMBNNs is provided to solve the MIDM through the parameter used in Tab. 1. The systems training is directed with the LMB scheme based back-propagation is oppressed to adjust the Jacobian presentation 'jZ', i.e., bias form of the variables Z and MSE by regulating the weight. The adjustment or adaptation of each variable of the decision through the LMB is shown as:

$$\begin{aligned} je &= jZ \times E, \\ jj &= jZ \times jZ, \\ dZ &= \frac{-(I \times mu + jj)}{je}, \end{aligned}$$

where I and E represent the unit matrix and error. Tab. 1 indicates the setting of the parameter based on the LMB process as well as minor change/variation/alteration, which conclude the poor presentation, i.e., precipitate convergence. Consequently, these situations will be combined with widespread care, after directing comprehensive numerical practice and investigation. Moreover, few necessary theoretical details based on the LMB algorithms along with the process of execution can be reported in the references [54-57].

4. Simulations with discussion and investigation

In this section, the numerical procedures for the solutions of three different variants of the FO-MIDM by applying the LMBNNs. The numerical results based on the FO-MIDM have been presented by selecting the values of FO (α), while, the other values are $\mu = 0.2$, $\beta = 0.1$, $\sigma = 0.5$, $\gamma = 0.3$, $\psi = 0.45$, $\delta = 0.03$ and $\alpha = 0.4$. The ICs are $m_1 = 0.1$, $m_2 = 0.2$, $m_3 = 0.3$ and $m_4 = 0.4$.

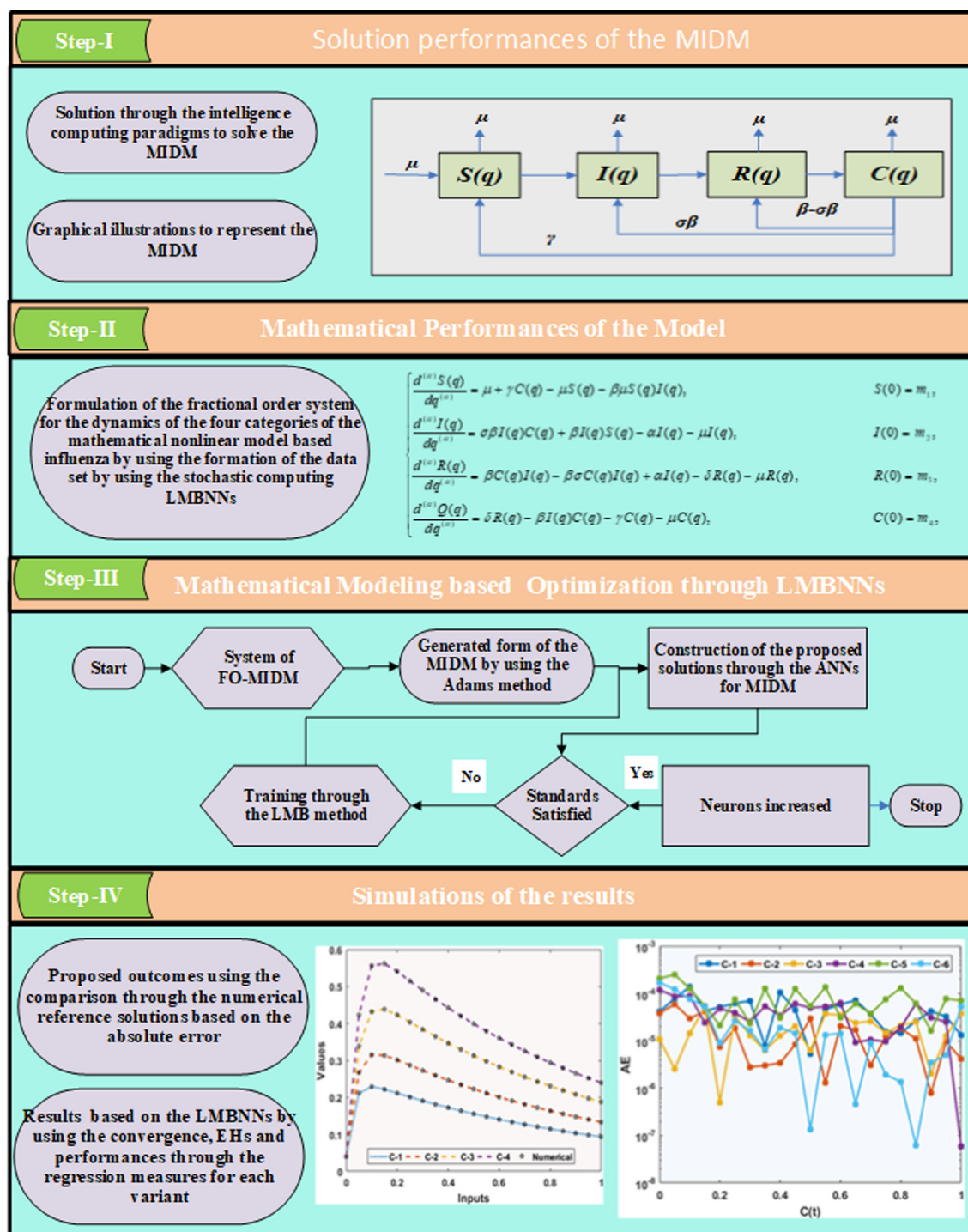
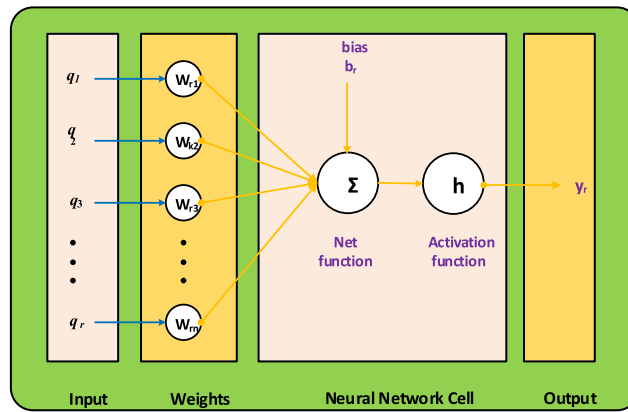
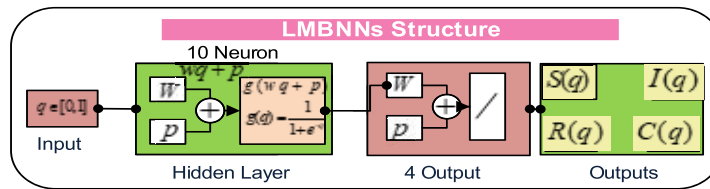


Fig. 2 Workflow representations of the designed LMBNNs for solving the FO-MIDM: Phase 1 shows the overview of the problem. Phase 2 represents the mathematical system illustration. Phase 3 designates the practical block using the LMBNNs. Phase 4 is the simulation outcomes along with analysis.



(a) Construction of general form of the single neuron



(b) Design structure of the layers, a single representation of the vector using the performances of input layer, hidden layer: sixteen neurons and output layer: four vectors

Fig. 3 Precise and generic construction of the ANNs for solving the FO-MIDM.

Case 1. Suppose the FO-MIDM is written as:

$$\begin{cases} \frac{d^{0.6}S(q)}{dq^{0.6}} = 0.2 + 0.3C(q) - 0.2S(q) - 0.02S(q)I(q), & S(0) = 0.1, \\ \frac{d^{0.6}I(q)}{dq^{0.6}} = 0.05I(q)C(q) + 0.1I(q)S(q) - 0.65I(q), & I(0) = 0.2, \\ \frac{d^{0.6}R(q)}{dq^{0.6}} = 0.1C(q)I(q) - 0.05C(q)I(q) + 0.45I(q) - 0.23R(q), & R(0) = 0.3, \\ \frac{d^{0.6}C(q)}{dq^{0.6}} = 0.03R(q) - 0.1I(q)C(q) - 0.3C(q), & C(0) = 0.4, \end{cases} \quad (3)$$

Case 2. Suppose the FO-MIDM is presented as:

$$\begin{cases} \frac{d^{0.7}S(q)}{dq^{0.7}} = 0.2 + 0.3C(q) - 0.2S(q) - 0.02S(q)I(q), & S(0) = 0.1, \\ \frac{d^{0.7}I(q)}{dq^{0.7}} = 0.05I(q)C(q) + 0.1I(q)S(q) - 0.65I(q), & I(0) = 0.2, \\ \frac{d^{0.7}R(q)}{dq^{0.7}} = 0.1C(q)I(q) - 0.05C(q)I(q) + 0.45I(q) - 0.23R(q), & R(0) = 0.3, \\ \frac{d^{0.7}C(q)}{dq^{0.7}} = 0.03R(q) - 0.1I(q)C(q) - 0.3C(q), & C(0) = 0.4, \end{cases} \quad (4)$$

Case 3. Suppose the FO-MIDM is shown as:

$$\begin{cases} \frac{d^{0.8}S(q)}{dq^{0.8}} = 0.2 + 0.3C(q) - 0.2S(q) - 0.02S(q)I(q), & S(0) = 0.1, \\ \frac{d^{0.8}I(q)}{dq^{0.8}} = 0.05I(q)C(q) + 0.1I(q)S(q) - 0.65I(q), & I(0) = 0.2, \\ \frac{d^{0.8}R(q)}{dq^{0.8}} = 0.1C(q)I(q) - 0.05C(q)I(q) + 0.45I(q) - 0.23R(q), & R(0) = 0.3, \\ \frac{d^{0.8}C(q)}{dq^{0.8}} = 0.03R(q) - 0.1I(q)C(q) - 0.3C(q), & C(0) = 0.4, \end{cases} \quad (5)$$

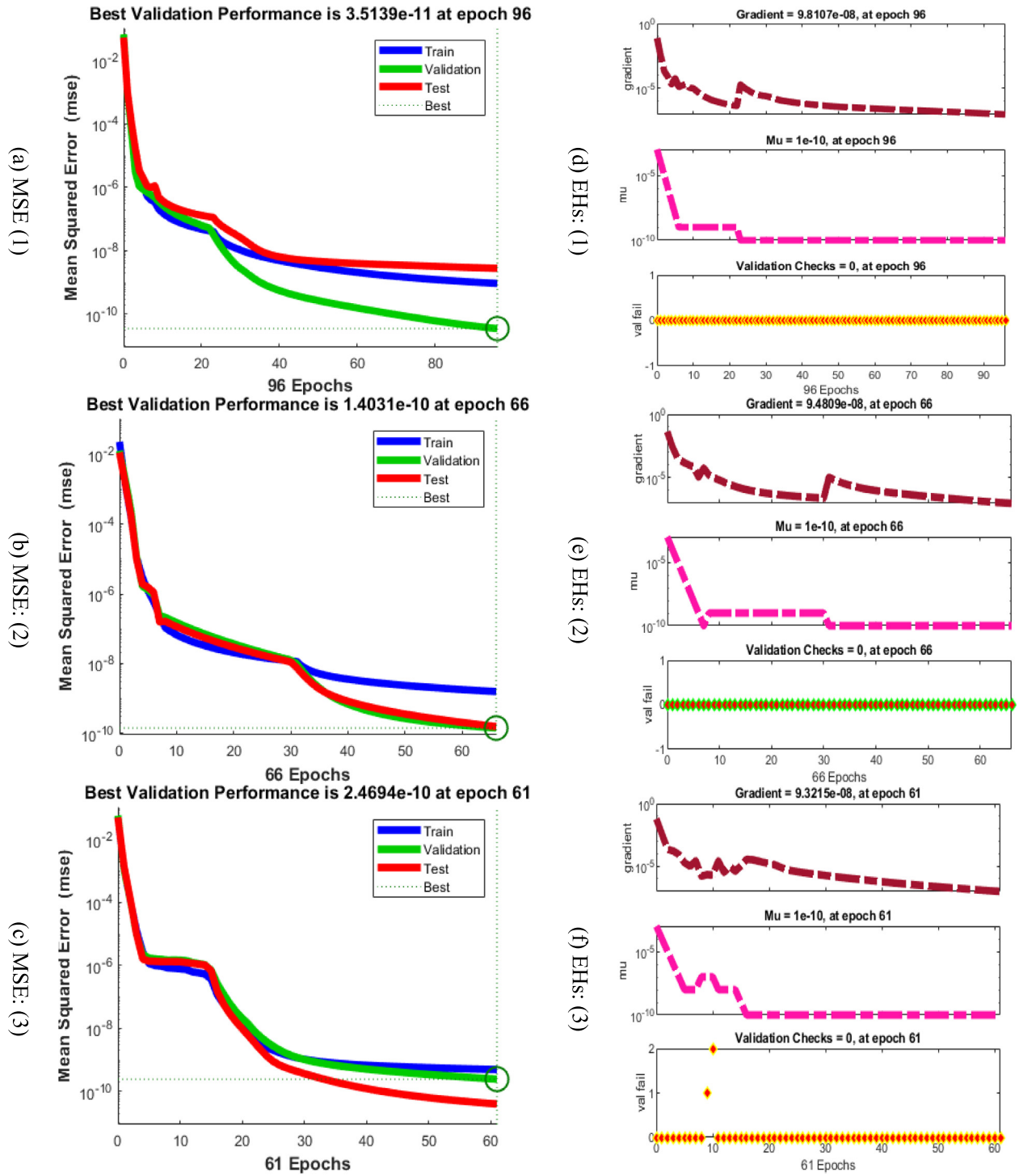


Fig. 4 MSE and STs measures for the FO-MIDM.

The achieved performances through the stochastic structure LMBNNs have been investigated with inputs $[0, 1]$ for the MIDM with 15 neurons together with the statics of training is selected as 78%, while for both authentication and testing measures have been chosen as 11%. The different values of these samples can also be selected. If the selection of the training values is $> 78\%$, then the better performances can be achieved because of the bias input values. Moreover, the data

for the training samples is $< 78\%$, then the accurateness of the LMBNNs is degraded considerably for the FO-MIDM. Hence, the sample data for the inputs of unbiased and bias should be adjusted with concentration to evade both divergence and premature convergence.

The numerical simulations of the FO-MIDM have been plotted in Figs. 4-6 by using the computational performances of the LMBNNs. Fig. 4 presents the performances of MSE

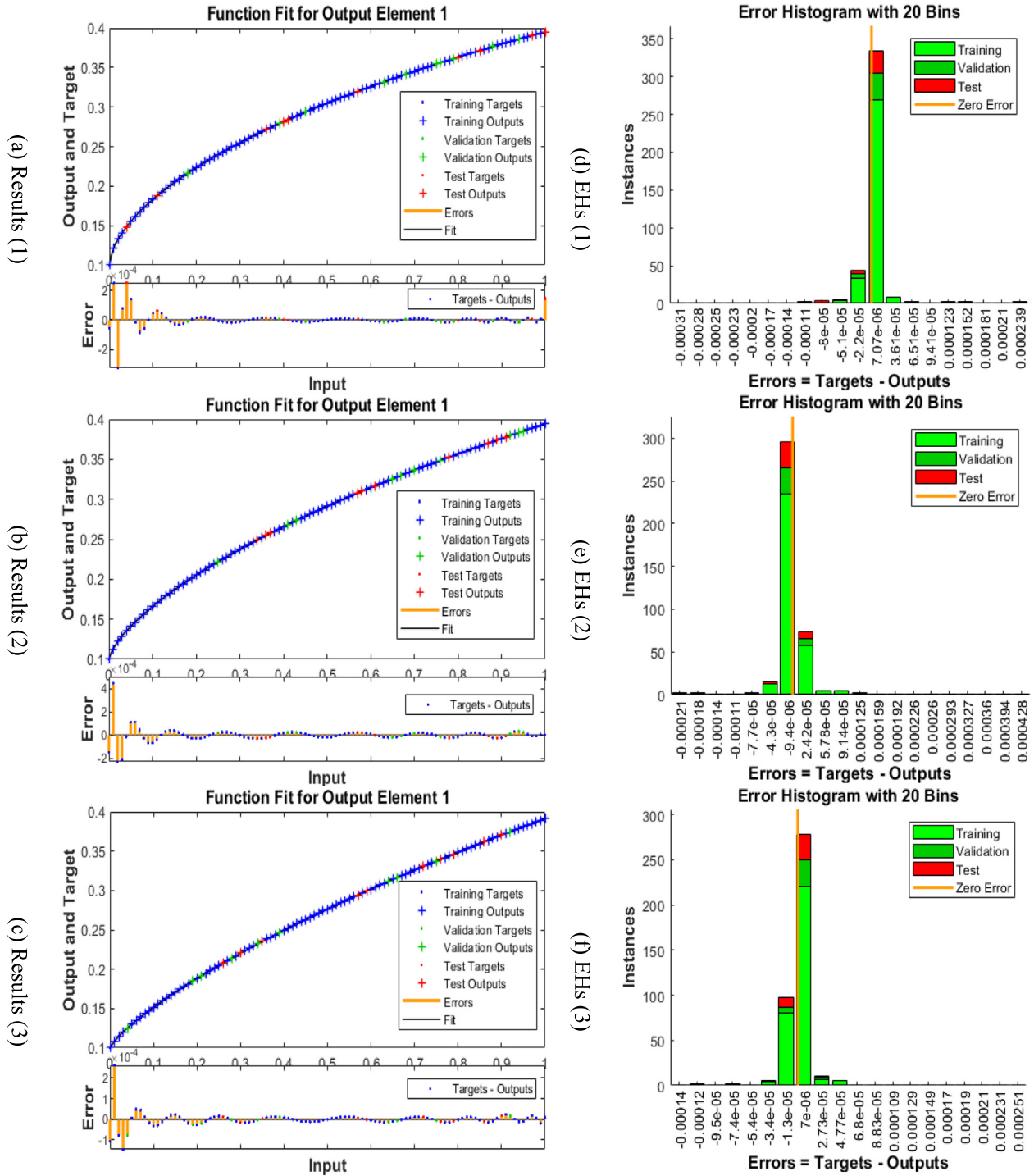
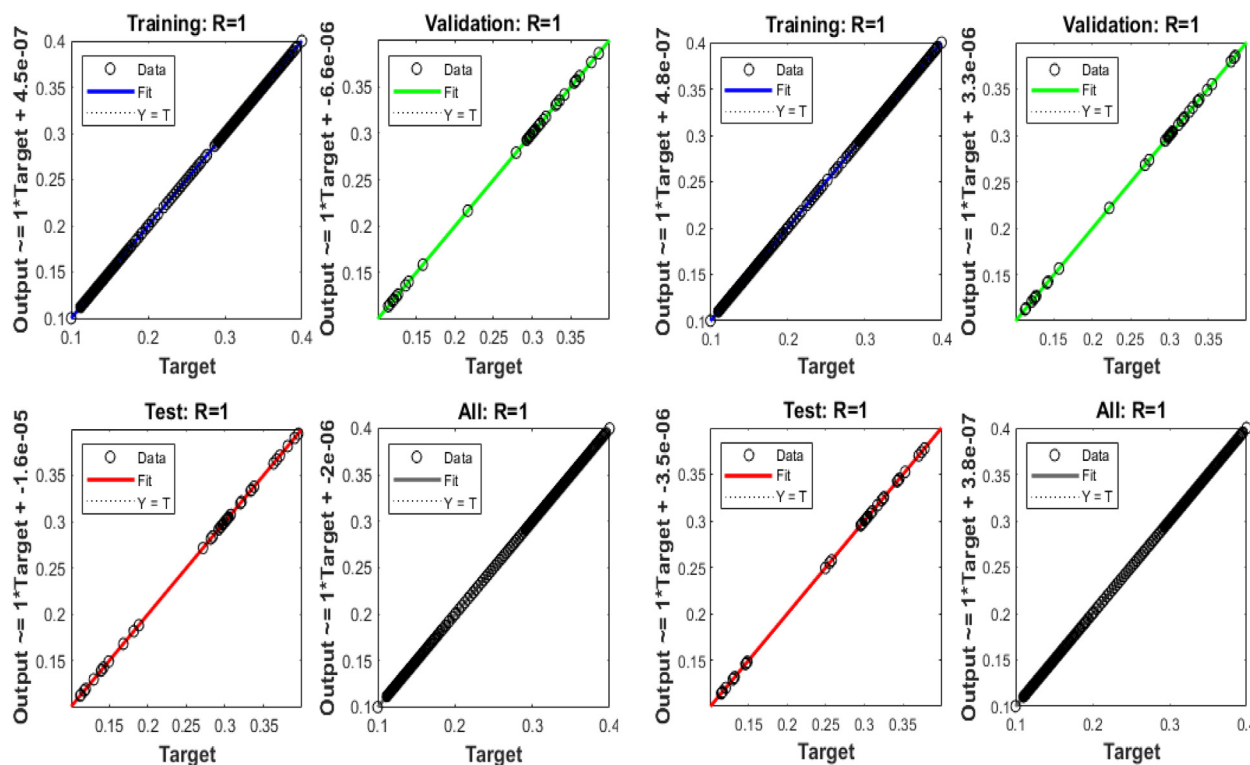


Fig. 5 Valuations of the results and EHs for the FO-MIDM.

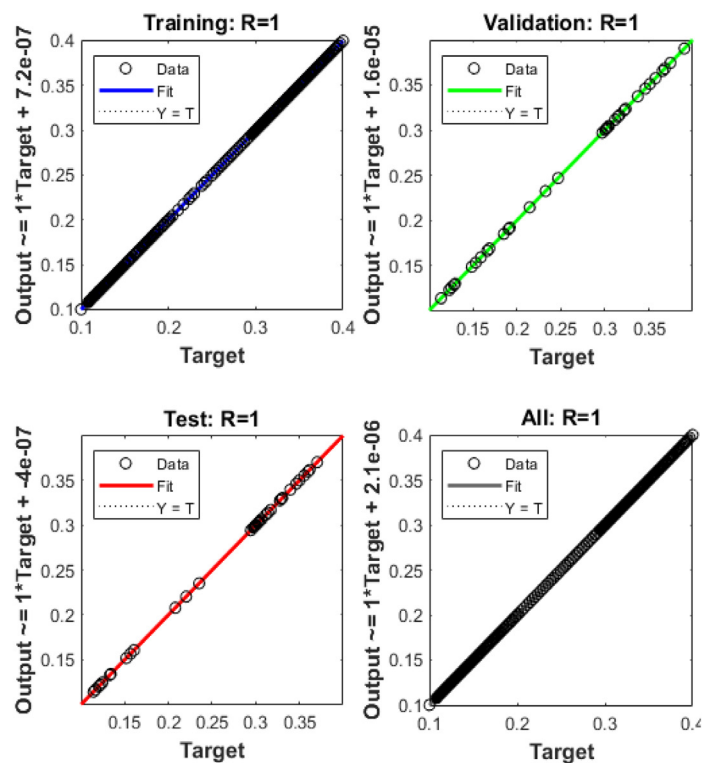
for the FO-MIDM. The best values of the obtained performances of the FO-MIDM have been calculated for FO case 1, 2 and 3 are 3.51×10^{-11} , 1.40×10^{-10} and 2.46×10^{-10} with Epochs 96, 66, and 61. The values of the gradient have been calculated 9.81×10^{-08} , 9.48×10^{-08} , and 9.32×10^{-08} for 1st, 2nd and 3rd variation. The obtained results and the EHs performances for the FO-MIDM using the LMBNNs are defined in Fig. 5. The EHs measures for of the FO-MIDM

have been calculated as 7.07×10^{-06} , -9.4×10^{-06} , and 7.0×10^{-06} . The best form of the testing, training and validation for the FO-MIM has also provided in Fig. 4. The values of the correlation have been presented in Fig. 6 using the computational procedure based LMBNNs for the FO-MIDM. Each variation of the FO-MIDM designated the correlation values 1 that signify the perfect model. The exactness of the scheme to solve the fractional kind of MIDM is described



(a) Regression (1)

(b) Regression (2)



(c) Regression (3)

Fig. 6 Regression for the FO-MIDM.

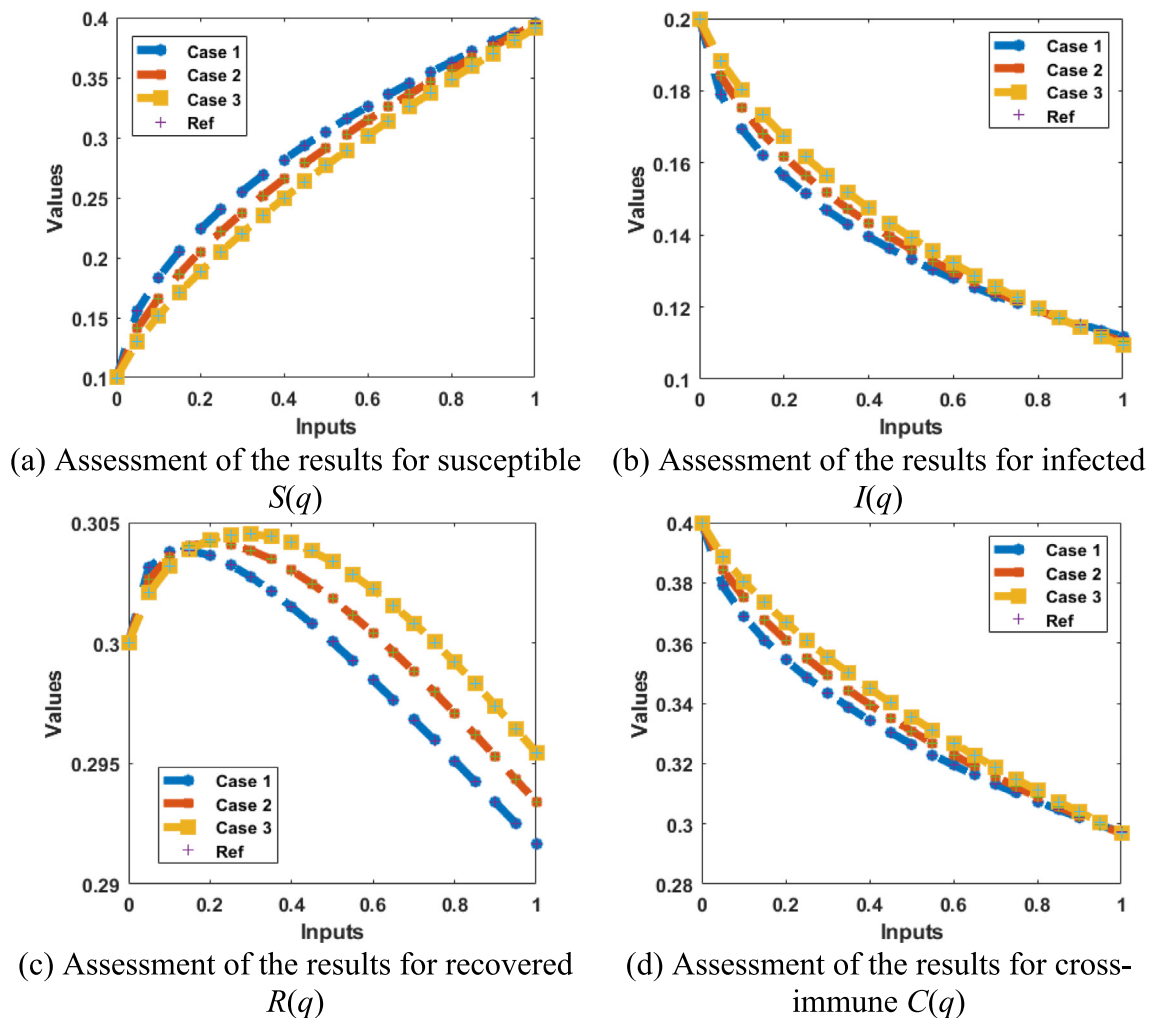


Fig. 7 Assessments of the results through the LMBNNs for solving the MIDM.

through the process of testing, validation and training. Tab. 2 performs the MSE values for the FO-MIDM using the stochastic procedure.

Figs. 7 and 8 provide the results comparison and AE values to solve the FO-MIDM. The comparison of the obtained and reference solutions has been presented in Fig. 7. The efficiency of the stochastic scheme is observed through the overlapping of the results (reference and obtained). These precise matchings of the results indicate the correctness and accuracy of the proposed stochastic structure for solving each category of the FO-MIDM. The performances of the AE using the proposed technique for solving the FO-MIDM have been presented in Fig. 8. These values based on the AE for the susceptible $S(q)$ class have been provided in Fig. 8(a) and calculated as 10^{-04} to 10^{-06} , 10^{-05} to 10^{-06} and 10^{-05} to 10^{-07} . For infected $I(q)$, the AE have been performed around 10^{-05} to 10^{-06} , 10^{-04} to 10^{-07} and 10^{-05} to 10^{-07} . The performances of the AE values for recovered $R(q)$ have been provided in Fig. 8(c) that are achieved around 10^{-05} to 10^{-06} , 10^{-05} to 10^{-08} and 10^{-05} to 10^{-07} . The AE for the last class cross-

immune people $C(q)$ have been calculate as 10^{-05} - 10^{-08} , 10^{-04} to 10^{-07} and 10^{-05} to 10^{-07} . The calculated AE measures present the exactness of the stochastic procedure LMBNNs for the FO-MIDM.

5. Conclusions

The aim of this research work is to provide the numerical simulations of the fractional order mathematical influenza disease nonlinear model by providing the computational context based on the stochastic Levenberg-Marquardt backpropagation neural networks. The fractional order investigations have been presented to achieve more precise performances of the MIDM as compared to the integer order. The solution of the three different variants using the FO derivatives has been provided to present the numerical simulations of each category of the MIDM. Fifteen neurons together with the statics of training have been selected as 78%, while for both authentication and testing measures have been chosen as 11%. For the

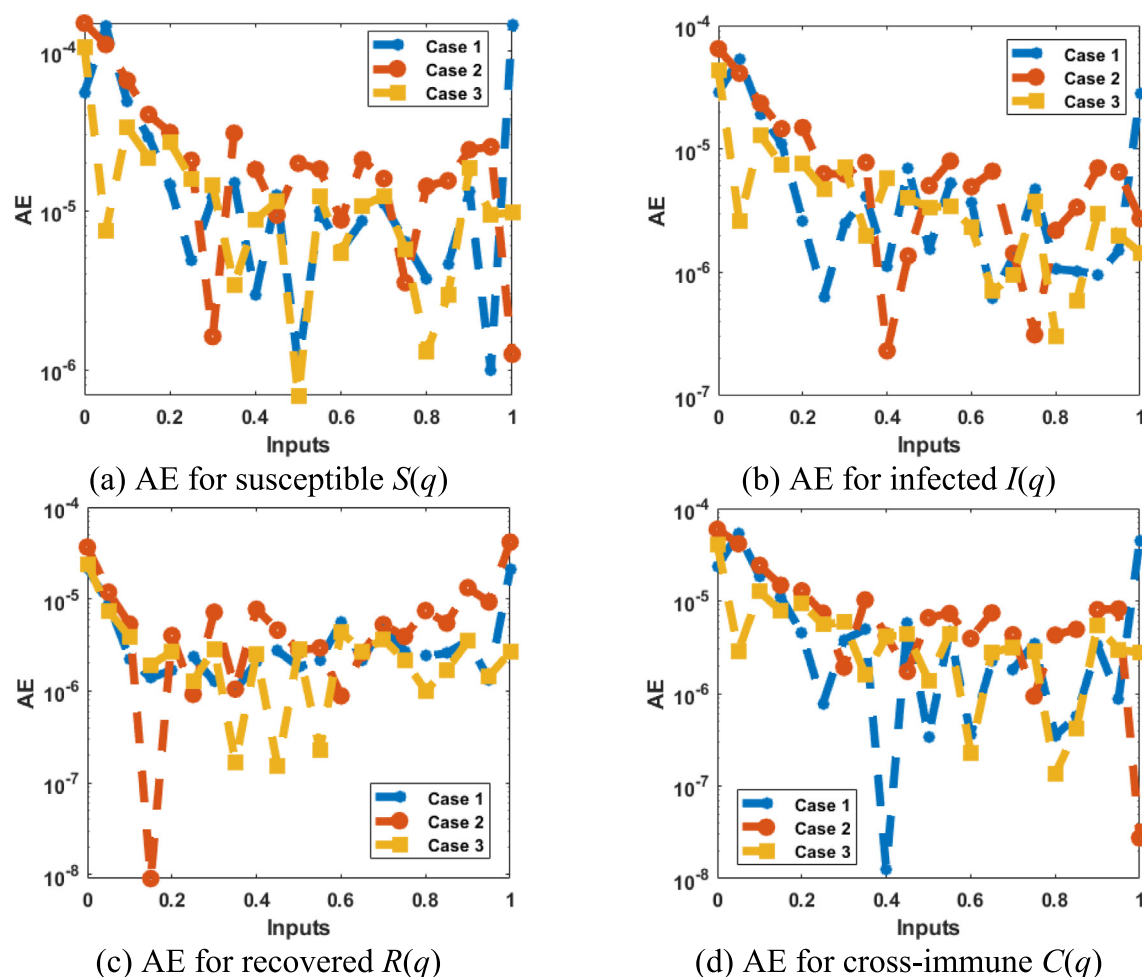


Fig. 8 AE performances through the LMBNNs for solving the MIDM.

competence, precision, capability and aptitude of the proposed computing structure, a comprehensive investigation has been accessed to perform the correlation, MSE, EHs, information of STs and regression analysis. The AE has been performed in good measures to solve the FO-MIDM. The values and importance of the LMBNNs procedure have been validated through the overlapping of the results with good measures up to the accuracy of 5 to 7 decimals for solving the MIDM.

In upcoming work, the proposed LMBNNs structure can be discovered in the system of the fluid dynamics, biological systems, wave solution models and nonlinear dynamical models [58–66].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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